Decision-Making Under Uncertainty: Exam Solutions

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Learning objective

Apply minimax regret and Expected Monetary Value (EMV) to select an alternative under uncertainty, and use sensitivity analysis to determine how a subjective probability changes the preferred alternative.

Criteria assessment

This task sheet assesses Criteria C5 through C7. Each problem assesses the criterion indicated in its title, and each problem is worth 2 points. Partial credit may be awarded when the correct procedure is clearly shown but arithmetic errors are made. Answers that do not include the required procedure will not receive credit. Failure to write your name on the task sheet or submitting it after the deadline will result in a one-point deduction from each criterion.

A minimum of 4 points is required to pass each criterion.

Evaluated criteria

C5 Applies the Minimax Regret criterion by constructing an opportunity-loss table and selecting the alternative with the smallest maximum regret.

C6 Calculates EMV using given probabilities and selects the alternative with the highest expected payoff.

C7 Uses subjective probabilities and sensitivity analysis to identify threshold probabilities and the preferred alternative over each feasible range.

1 Community Garden Greenhouse [C5-2]

A community garden must choose one of three greenhouse plans. Annual net benefits (in thousands of dollars) depend on the growing season.

Alternative	Excellent (S_1)	Average (S_2)	Poor (S_3)
Large greenhouse (A_1)	70	30	-10
Medium greenhouse (A_2)	55	45	20
Small greenhouse (A_3)	35	40	32

Use the Minimax Regret criterion to determine which plan should be selected. Show the best payoff in each state, the regret table, and each alternative's maximum regret. **[2]**

C5 — Solution

The best payoff in each state is

$$b(S_1) = \max\{70, 55, 35\} = 70, \quad b(S_2) = \max\{30, 45, 40\} = 45, \quad b(S_3) = \max\{-10, 20, 32\} = 32.$$

Using $r(A_i, S_j) = b(S_j) - P_{ay}(A_i, S_j)$ gives

	S_1	S_2	S_3	Maximum regret
A_1	$70 - 70 = 0$	$45 - 30 = 15$	$32 - (-10) = 42$	42
A_2	$70 - 55 = 15$	$45 - 45 = 0$	$32 - 20 = 12$	15
A_3	$70 - 35 = 35$	$45 - 40 = 5$	$32 - 32 = 0$	35

Thus $\min\{42, 15, 35\} = 15$, so the Minimax Regret criterion selects A_2, the medium greenhouse.



2 Coastal Ferry Capacity [C5-2]

A ferry operator is choosing a capacity plan. Seasonal profit (in thousands of dollars) is shown below; no reliable probabilities are available.

Alternative	High demand (S_1)	Normal demand (S_2)	Low demand (S_3)
Add two ferries (A_1)	100	35	-25
Add one ferry (A_2)	75	55	10
Lease peak capacity (A_3)	55	48	30
Keep current fleet (A_4)	30	30	30

Determine the decision using Minimax Regret. **[2]**

C5 — Solution

The statewise best payoffs are $b(S_1) = 100$, $b(S_2) = 55$, and $b(S_3) = 30$.

	S_1	S_2	S_3	Maximum regret
A_1	$100 - 100 = 0$	$55 - 35 = 20$	$30 - (-25) = 55$	55
A_2	$100 - 75 = 25$	$55 - 55 = 0$	$30 - 10 = 20$	25
A_3	$100 - 55 = 45$	$55 - 48 = 7$	$30 - 30 = 0$	45
A_4	$100 - 30 = 70$	$55 - 30 = 25$	$30 - 30 = 0$	70

Since $\min\{55, 25, 45, 70\} = 25$, the recommended decision is A_2 : add one ferry.



3 Mobile Clinic Schedule [C5-2]

A health organization must select a mobile-clinic schedule. Net community value (in thousands of dollars) depends on attendance.

Alternative	Very high (S_1)	High (S_2)	Moderate (S_3)	Low (S_4)
Full schedule (A_1)	90	50	20	-20
Flexible schedule (A_2)	70	60	45	10
Limited schedule (A_3)	45	48	42	35

Use Minimax Regret to select a schedule. **[2]**

C5 — Solution

The best-payoff vector is

$$(b(S_1), b(S_2), b(S_3), b(S_4)) = (90, 60, 45, 35).$$

	S_1	S_2	S_3	S_4	Maximum regret
A_1	0	10	25	55	55
A_2	20	0	0	25	25
A_3	45	12	3	0	45

For example, $r(A_1, S_4) = 35 - (-20) = 55$ and $r(A_3, S_2) = 60 - 48 = 12$. The smallest maximum regret is $\min\{55, 25, 45\} = 25$, so select A_2, the flexible schedule.



4 Urban Farm Crop Mix [C6-2]

An urban farm is selecting a crop mix. Profits are in thousands of dollars.

	Strong market (S_1)	Stable market (S_2)	Weak market (S_3)
$P(S_j)$	0.50	0.30	0.20
Specialty crops (A_1)	80	40	-10
Balanced crops (A_2)	60	55	20
Staple crops (A_3)	40	45	35

Calculate every EMV and apply the Bayes decision criterion. **[2]**

C6 — Solution

First, $0.50 + 0.30 + 0.20 = 1$. Then

$$EMV(A_1) = (0.50)(80) + (0.30)(40) + (0.20)(-10) = 40 + 12 - 2 = 50,$$
$$EMV(A_2) = (0.50)(60) + (0.30)(55) + (0.20)(20) = 30 + 16.5 + 4 = 50.5,$$
$$EMV(A_3) = (0.50)(40) + (0.30)(45) + (0.20)(35) = 20 + 13.5 + 7 = 40.5.$$

The highest EMV is $\max\{50, 50.5, 40.5\} = 50.5$. Therefore, choose A_2, the balanced crop mix, with expected profit \$50,500.



5 Museum Exhibition Scale [C6-2]

A museum is deciding the scale of a temporary exhibition. Payoffs are in thousands of dollars.

	High attendance (S_1)	Expected attendance (S_2)	Low attendance (S_3)
$P(S_j)$	0.25	0.45	0.30
Major exhibition (A_1)	95	35	-20
Standard exhibition (A_2)	70	50	15
Compact exhibition (A_3)	50	45	32
Permanent collection only (A_4)	25	25	25

Calculate the EMVs and recommend an alternative. **[2]**

C6 — Solution

The probabilities are valid because $0.25 + 0.45 + 0.30 = 1$.

$$EMV(A_1) = (0.25)(95) + (0.45)(35) + (0.30)(-20) = 23.75 + 15.75 - 6 = 33.50,$$
$$EMV(A_2) = (0.25)(70) + (0.45)(50) + (0.30)(15) = 17.5 + 22.5 + 4.5 = 44.50,$$
$$EMV(A_3) = (0.25)(50) + (0.45)(45) + (0.30)(32) = 12.5 + 20.25 + 9.6 = 42.35,$$
$$EMV(A_4) = (0.25)(25) + (0.45)(25) + (0.30)(25) = 25.00.$$

Because 44.50 is the highest EMV, the Bayes decision is A_2, the standard exhibition, with expected payoff \$44,500.



6 Solar Installer Staffing [C6-2]

A solar installer is choosing a staffing plan for next quarter. Payoffs are in thousands of dollars.

	Surge (S_1)	Strong (S_2)	Steady (S_3)	Slow (S_4)
$P(S_j)$	0.15	0.35	0.30	0.20
Rapid expansion (A_1)	100	60	20	-30
Moderate expansion (A_2)	75	65	45	10
Current staff (A_3)	50	52	48	40

Use EMV to determine the preferred staffing plan. **[2]**

C6 — Solution

The probability check is $0.15 + 0.35 + 0.30 + 0.20 = 1$.

$$EMV(A_1) = (0.15)(100) + (0.35)(60) + (0.30)(20) + (0.20)(-30) = 15 + 21 + 6 - 6 = 36,$$
$$EMV(A_2) = (0.15)(75) + (0.35)(65) + (0.30)(45) + (0.20)(10) = 11.25 + 22.75 + 13.5 + 2 = 49.5,$$
$$EMV(A_3) = (0.15)(50) + (0.35)(52) + (0.30)(48) + (0.20)(40) = 7.5 + 18.2 + 14.4 + 8 = 48.1.$$

Thus $\max\{36, 49.5, 48.1\} = 49.5$, so choose A_2, moderate expansion, with expected payoff \$49,500.



7 Festival Catering Contract [C7-2]

A caterer must choose between a large contract (A_1) and a small contract (A_2). Payoffs (in thousands of dollars) depend on whether festival attendance is high (S_1) or low (S_2).

Alternative	High (S_1)	Low (S_2)
Large contract (A_1)	80	10
Small contract (A_2)	45	35

Let $P(S_1) = p$ and $P(S_2) = 1 - p$. Find the threshold probability and state the preferred alternative for every $0 \le p \le 1$. **[2]**

C7 — Solution

$$EMV(A_1) = 80p + 10(1-p) = 10 + 70p,$$
$$EMV(A_2) = 45p + 35(1-p) = 35 + 10p.$$

At the threshold, $10 + 70p = 35 + 10p$, so $60p = 25$ and $p = \frac{5}{12} \approx 0.4167$. Since A_2 has the higher EMV at $p = 0$ and A_1 has the higher EMV at $p = 1$,

$$\begin{cases} A_2, & 0 \le p < \frac{5}{12}, \\ A_1 \text{ and } A_2 \text{ are tied}, & p = \frac{5}{12}, \\ A_1, & \frac{5}{12} < p \le 1. \end{cases}$$



8 Electric-Bike Launch [C7-2]

A retailer is comparing an aggressive launch (A_1) with a cautious launch (A_2). Payoffs are in thousands of dollars.

Alternative	High demand (S_1)	Moderate demand (S_2)	Low demand (S_3)
Aggressive launch (A_1)	90	30	-10
Cautious launch (A_2)	45	50	25

Suppose $P(S_1) = p$ and $P(S_2) = \frac{p}{2}$. Determine the feasible range of p, the threshold, and the preferred launch over the entire range. **[2]**

C7 — Solution

Because the three state probabilities must sum to one,

$$P(S_3) = 1 - p - \frac{p}{2} = 1 - \frac{3p}{2}.$$

Nonnegative probabilities require $p \ge 0$ and $1 - \frac{3p}{2} \ge 0$, so the feasible range is $0 \le p \le \frac{2}{3}$.

$$EMV(A_1) = 90p + 30\left(\frac{p}{2}\right) - 10\left(1 - \frac{3p}{2}\right) = 120p - 10,$$
$$EMV(A_2) = 45p + 50\left(\frac{p}{2}\right) + 25\left(1 - \frac{3p}{2}\right) = 32.5p + 25.$$

Equating the EMVs gives $120p - 10 = 32.5p + 25$, hence

$$87.5p = 35 \quad \Rightarrow \quad p = \frac{2}{5} = 0.4.$$

Therefore,

$$\begin{cases} A_2, & 0 \le p < \frac{2}{5}, \\ A_1 \text{ and } A_2 \text{ are tied}, & p = \frac{2}{5}, \\ A_1, & \frac{2}{5} < p \le \frac{2}{3}. \end{cases}$$



9 Rainwater-System Bid [C7-2]

An engineering firm must choose between a standard bid (A_1) and an innovative bid (A_2). Payoffs are in thousands of dollars.

Alternative	Immediate approval (S_1)	Revision (S_2)	Delayed approval (S_3)
Standard bid (A_1)	20	55	75
Innovative bid (A_2)	70	40	30

Suppose $P(S_1) = 2p$ and $P(S_2) = p$. Find the feasible range, threshold probability, and preferred bid for each range of p. **[2]**

C7 — Solution

The remaining probability is

$$P(S_3) = 1 - 2p - p = 1 - 3p.$$

The probability conditions $p \ge 0$ and $1 - 3p \ge 0$ give the feasible range $0 \le p \le \frac{1}{3}$.

$$EMV(A_1) = 20(2p) + 55p + 75(1 - 3p) = 75 - 130p,$$
$$EMV(A_2) = 70(2p) + 40p + 30(1 - 3p) = 30 + 90p.$$

At indifference,

$$75 - 130p = 30 + 90p \quad \Rightarrow \quad 45 = 220p \quad \Rightarrow \quad p = \frac{9}{44} \approx 0.2045.$$

Here A_1 is preferred below the threshold, while the faster-growing EMV of A_2 makes it preferred above the threshold:

$$\begin{cases} A_1, & 0 \le p < \frac{9}{44}, \\ A_1 \text{ and } A_2 \text{ are tied}, & p = \frac{9}{44}, \\ A_2, & \frac{9}{44} < p \le \frac{1}{3}. \end{cases}$$



10 Summarized Solutions

10.1 Community Garden Greenhouse C5–2

The best-payoff vector is $(70, 45, 32)$. The regret table and row maxima are

	S_1	S_2	S_3	$\max r$
A_1	$70-70$ $=0$	$45-30$ $=15$	$32-(-10)$ $=42$	$\max\{0,15,42\}$ $=42$
A_2	$70-55$ $=15$	$45-45$ $=0$	$32-20$ $=12$	$\max\{15,0,12\}$ $=15$
A_3	$70-35$ $=35$	$45-40$ $=5$	$32-32$ $=0$	$\max\{35,5,0\}$ $=35$

Thus $\min\{42, 15, 35\} = 15$, so select A_2, the medium greenhouse.

10.2 Coastal Ferry Capacity C5–2

The best-payoff vector is $(100, 55, 30)$. The regret table and row maxima are

	S_1	S_2	S_3	$\max r$
A_1	$100-100$ $=0$	$55-35$ $=20$	$30-(-25)$ $=55$	$\max\{0,20,55\}$ $=55$
A_2	$100-75$ $=25$	$55-55$ $=0$	$30-10$ $=20$	$\max\{25,0,20\}$ $=25$
A_3	$100-55$ $=45$	$55-48$ $=7$	$30-30$ $=0$	$\max\{45,7,0\}$ $=45$
A_4	$100-30$ $=70$	$55-30$ $=25$	$30-30$ $=0$	$\max\{70,25,0\}$ $=70$

Thus $\min\{55, 25, 45, 70\} = 25$, so select A_2, add one ferry.

10.3 Mobile Clinic Schedule C5–2

The best-payoff vector is $(90, 60, 45, 35)$. The regret table and row maxima are

	S_1	S_2	S_3	S_4	$\max r$
A_1	$90-90$ $=0$	$60-50$ $=10$	$45-20$ $=25$	$35-(-20)$ $=55$	$\max\{0,10,25,55\}$ $=55$
A_2	$90-70$ $=20$	$60-60$ $=0$	$45-45$ $=0$	$35-10$ $=25$	$\max\{20,0,0,25\}$ $=25$
A_3	$90-45$ $=45$	$60-48$ $=12$	$45-42$ $=3$	$35-35$ $=0$	$\max\{45,12,3,0\}$ $=45$

Thus $\min\{55, 25, 45\} = 25$, so select A_2, the flexible schedule.

10.4 Urban Farm Crop Mix C6–2

Since $0.50 + 0.30 + 0.20 = 1$,

$$EMV(A_1) = (0.50)(80) + (0.30)(40) + (0.20)(-10) = 50,$$
$$EMV(A_2) = (0.50)(60) + (0.30)(55) + (0.20)(20) = 50.5,$$
$$EMV(A_3) = (0.50)(40) + (0.30)(45) + (0.20)(35) = 40.5.$$

The largest EMV is 50.5, so select A_2, the balanced crop mix ($50,500).

10.5 Museum Exhibition Scale C6–2

Since $0.25 + 0.45 + 0.30 = 1$,

$$EMV(A_1) = (0.25)(95) + (0.45)(35) + (0.30)(-20) = 33.50,$$
$$EMV(A_2) = (0.25)(70) + (0.45)(50) + (0.30)(15) = 44.50,$$
$$EMV(A_3) = (0.25)(50) + (0.45)(45) + (0.30)(32) = 42.35,$$
$$EMV(A_4) = 25.00.$$

The largest EMV is 44.50, so select A_2, the standard exhibition ($44,500).

10.6 Solar Installer Staffing C6–2

Since $0.15 + 0.35 + 0.30 + 0.20 = 1$,

$$EMV(A_1) = (0.15)(100) + (0.35)(60) + (0.30)(20) + (0.20)(-30) = 36,$$
$$EMV(A_2) = (0.15)(75) + (0.35)(65) + (0.30)(45) + (0.20)(10) = 49.5,$$
$$EMV(A_3) = (0.15)(50) + (0.35)(52) + (0.30)(48) + (0.20)(40) = 48.1.$$

The largest EMV is 49.5, so select A_2, moderate expansion ($49,500).

10.7 Festival Catering Contract C7–2

$$EMV(A_1) = 10 + 70p, \qquad EMV(A_2) = 35 + 10p.$$

Indifference requires $10 + 70p = 35 + 10p$, so $p = \frac{5}{12} \approx 0.4167$. Therefore

$$A_2 \ (0 \le p < \tfrac{5}{12}); \quad A_1 = A_2 \ (p = \tfrac{5}{12}); \quad A_1 \ (\tfrac{5}{12} < p \le 1).$$

10.8 Electric-Bike Launch C7–2

Here $P(S_3) = 1 - \frac{3p}{2}$, so $0 \le p \le \frac{2}{3}$. Also,

$$EMV(A_1) = 120p - 10, \qquad EMV(A_2) = 32.5p + 25.$$

Indifference requires $120p - 10 = 32.5p + 25$, so $p = \frac{2}{5}$. Therefore

$$A_2 \ (0 \le p < \tfrac{2}{5}); \quad A_1 = A_2 \ (p = \tfrac{2}{5}); \quad A_1 \ (\tfrac{2}{5} < p \le \tfrac{2}{3}).$$

Here $P(S_3) = 1 - 3p$, so $0 \le p \le \frac{1}{3}$. Also,

$$EMV(A_1) = 75 - 130p, \qquad EMV(A_2) = 30 + 90p.$$

Indifference requires $75 - 130p = 30 + 90p$, so $p = \frac{9}{44} \approx 0.2045$. Therefore

$$A_1 \ (0 \le p < \tfrac{9}{44}); \quad A_1 = A_2 \ (p = \tfrac{9}{44}); \quad A_2 \ (\tfrac{9}{44} < p \le \tfrac{1}{3}).$$